POSITIVE LEBESGUE MEASURE OF AN ARITHMETIC SET FOR CONSTANT NON-CENTRAL SECTIONS

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Abstract. Haddad and Ryabogin proved that a symmetric convex body of revolution in $\mathbb{R}^4$ containing the unit ball, all of whose sections by hyperplanes tangent to the ball have the same volume A, is a Euclidean ball, provided that A belongs to an explicit set $\mathbb{A}$ defined by infinitely many Diophantine inequalities. They showed that $\mathbb{A}$ has positive Hausdorff dimension and asked whether it has positive Lebesgue measure. We answer this question affirmatively: every neighbourhood of their explicit example $A_0 = \frac{4\pi}{3}\tan\bigl(\pi(4+\sqrt{2})/14\bigr)^3$ contains a subset of $\mathbb{A}$ of positive Lebesgue measure. The proof replaces the parity conditions on continued fraction digits used by Haddad and Ryabogin with a direct estimate for the second smallest of the distances from the odd multiples of the rotation number to the nearest integers.

1. Introduction

Let B^n denote the centred unit Euclidean ball in $\mathbb{R}^n$. Barker and Larman [1] asked whether a convex body in $\mathbb{R}^n$ containing B^n in its interior is determined by the $(n-1)$-dimensional volumes of its sections by hyperplanes tangent to B^n. A particular case asks whether such a body, all of whose sections by tangent hyperplanes have the same volume, must be a Euclidean ball. The planar case goes back to Santaló [6], and Yaskin and Yaskina [7] settled the three-dimensional case under the additional assumption that the body has the volume of the corresponding ball. We refer to [2] for background on geometric tomography. The problem remains open in general.

Haddad and Ryabogin [4] treated dimension four for symmetric bodies of revolution, under an arithmetic condition on the common volume. Write $\omega_3 = 4\pi/3$ for the volume of B^3 and, for $A > 0$,

$$\gamma_A = \pi^{-1}\arctan\bigl((A/\omega_3)^{1/3}\bigr).$$

They introduced a set $\mathbb{A} \subseteq (8\omega_3, \infty)$, recalled in Definition 2.2, consisting of those A for which γ_A is irrational and which satisfy an infinite family of inequalities $C(A,k)$, $k \geq 2$, involving the distances from the odd multiples $(2k-1)\gamma_A$ to the nearest integers. They proved that for $A \in \mathbb{A}$ the answer to the question above is positive in the class of symmetric bodies of revolution in $\mathbb{R}^4$ [4, Theorem 9]; see Theorem 2.3 below. They showed that $\mathbb{A}$ contains the number

$$A_0 = \omega_3 \tan(\pi\gamma_0)^3, \qquad \gamma_0 = \frac{4+\sqrt{2}}{14}$$

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(numerically $A_0 \approx 81.5849$), that $\mathbb{A}$ is infinite and that it has positive Hausdorff dimension [4, Section 4.2], and they asked the following question [4, Question 15]: *Is $\mathbb{A}$ a set with positive Lebesgue measure?* Theorem 1.1 answers this question.

Theorem 1.1. *Every neighbourhood of A_0 contains a Borel subset of $\mathbb{A}$ of positive Lebesgue measure. In particular, $\mathbb{A}$ has positive Lebesgue measure.*

The set $\mathbb{A}$ is itself a Borel set (Lemma 2.4). Combined with [4, Theorem 9], Theorem 1.1 gives the following.

Corollary 1.2. *For every neighbourhood W of A_0 there is a Borel set $E \subseteq \mathbb{A} \cap W$ of positive Lebesgue measure with the following property. Let $A \in E$, and let $C \subseteq \mathbb{R}^4$ be a symmetric convex body of revolution containing B^4 such that $|C \cap H|_3 = A$ for every hyperplane H tangent to B^4. Then C is a Euclidean ball.*

Corollary 1.2 concerns only symmetric bodies of revolution in dimension four and the values $A \in \mathbb{A}$; it does not settle the Barker–Larman problem. In the opposite direction, Haddad [3] constructed approximate counterexamples near $\lambda_0 = 4\sqrt{3}\,\pi$, a value with $\gamma_{\lambda_0} = 1/3$; note that $\lambda_0 < 8\omega_3$. He also describes $\mathbb{A}$ as nowhere dense [3, Section 1]; this is consistent with Theorem 1.1, since a nowhere dense set can have positive Lebesgue measure.

Idea of the proof. As explained in [4, Section 4], the difficulty is that only odd multiples of γ_A enter the conditions $C(A,k)$. Haddad and Ryabogin therefore imposed parity conditions on the continued fraction digits of γ_A under which the denominators q_j, $j \geq 2$, of its convergents are odd [4, Proposition 13 and Theorem 14]; the resulting sets have Lebesgue measure zero [4, p. 19]. In their concluding remarks [4, Section 5] they observe that the parity conditions can be dropped by working with the subsequence of odd denominators, which is infinite because q_j is odd whenever q_{j-2} is odd and q_{j-1} is even, but that the resulting conditions become considerably more complicated, with bounds that are practically intractable, and they left Question 15 open.

We avoid odd denominators altogether. Passing to $\theta = 2\gamma_A$, we have $(2k-1)\gamma_A = (k-\frac{1}{2})\theta$. If p_n/q_n is any convergent of θ with $q_n \geq 2$, the points $(r-\frac{1}{2})\theta$, $1 \leq r \leq q_n$, lie within $1/q_{n+1}$ of a translate of the grid $q_n^{-1}\mathbb{Z}$ modulo 1, so the second smallest of their distances to the nearest integers is less than $3/q_n$ (Lemma 3.2). This bound holds for every convergent and does not depend on the parities of the denominators. Together with quadratic bounds for the functions in $C(A,k)$ (Lemma 3.1) and the classical lower bound for $\|m\theta\|$, it yields an explicit criterion: if the digits of θ satisfy $b_l \leq l^2$ for all $l \geq 257$, then $C(A,k)$ holds for all $k \geq q_{256}$ simultaneously (Proposition 4.2). Since this criterion imposes no parity condition, it is satisfied on a set of positive measure. More precisely, let $\theta_0 = 2\gamma_0$ and let I be the cylinder of numbers whose first $L \geq 256$ digits agree with those of θ_0. The fixed digits b_l, $257 \leq l \leq L$, are those of θ_0 and satisfy the criterion, while the condition $b_l \leq l^2$ for all $l > L$ fails only on a subset of I of relative measure at most $2/L$ (Lemma 5.1). The finitely many remaining conditions $C(A,k)$, $2 \leq k < q_{256}$, are open conditions that hold at A_0, so they persist in a neighbourhood of A_0, and L is chosen so that I lies in that neighbourhood.

Dependencies. Besides elementary facts on continued fractions [5], the proof of Theorem 1.1 uses one result of [4]: the membership $A_0 \in \mathbb{A}$ established in

[4, Section 4.2]. That membership rests on [4, Theorem 14(a)], whose hypothesis for A_0 reduces to the single inequality (7.1), and on the five initial conditions $C(A_0, k)$, $2 \le k \le 6$. In Section 7 we recall these and describe an independent check in exact rational interval arithmetic, contained in the archive `supplementary-certificate.zip` that accompanies this paper. Corollary 1.2 additionally uses [4, Theorem 9].

Organization. Section 2 recalls the definition of $\mathbb{A}$ and collects facts on continued fractions. Sections 3, 4 and 5 contain the basic estimates, the tail criterion and the measure estimate in cylinders. Section 6 proves Theorem 1.1 and Corollary 1.2, and Section 7 discusses the example A_0.

2. Preliminaries

2.1. **The set of Haddad and Ryabogin.** Throughout, $\|x\|$ denotes the distance from $x \in \mathbb{R}$ to the nearest integer, and $|X|$ (without subscript) the Lebesgue measure of a set $X \subseteq \mathbb{R}$. For $A > 8\omega_3$ put

$$R_A = \sec(\pi\gamma_A), \qquad \text{so that} \qquad A = \omega_3(R_A^2 - 1)^{3/2},$$

and for $x \ge 1$

$$G_A(x) = \arctan\sqrt{x^2-1} - \arctan\sqrt{R_A^2-1} = \arccos(1/x) - \pi\gamma_A.$$

For $0 < s < t < \pi/2$, and whenever $R_A \cos s \ge 1$, define

$$F_{A,-}(s) = -G_A(R_A\cos s), \qquad F_{A,+}(s,t) = G_A\Big(R_A\frac{\sin(t-s)}{\sin t - \sin s}\Big).$$

For $k \ge 1$ let $\delta_{A,k} = \pi\|(2k-1)\gamma_A\|$, and for $k \ge 2$ let $s_{A,k} < t_{A,k}$ be the two smallest elements of $\{\delta_{A,1}, \dots, \delta_{A,k}\}$. In [4] these are denoted $\alpha_{A,k}$ and $\beta_{A,k}$, and $F_{A,+}(\alpha,\beta)$ is written with $\sin(\alpha-\beta)/(\sin\alpha - \sin\beta)$, which has the same value. The next lemma shows that all these quantities are well defined when γ_A is irrational.

Lemma 2.1. *Let $A > 8\omega_3$ with γ_A irrational.*

1. *The numbers $\delta_{A,k}$, $k \ge 1$, are pairwise distinct and lie in $(0, \pi/2)$.*
2. *For every $k \ge 2$ we have $s_{A,k} \le \delta_{A,2} < \delta_{A,1} = \pi\gamma_A$, and hence $R_A\cos s_{A,k} > 1$.*
3. *For $0 < s < t < \pi/2$,*

$$\frac{\sin(t-s)}{\sin t - \sin s} = \frac{\cos\frac{t-s}{2}}{\cos\frac{t+s}{2}} = 1 + \frac{2\sin\frac{s}{2}\sin\frac{t}{2}}{\cos\frac{t+s}{2}} > 1. \tag{2.1}$$

Proof. (1) If $\|x\| = \|y\|$, then $x - y \in \mathbb{Z}$ or $x + y \in \mathbb{Z}$. For $x = (2k-1)\gamma_A$ and $y = (2k'-1)\gamma_A$ with $k \ne k'$, the numbers $x - y = 2(k-k')\gamma_A$ and $x + y = 2(k+k'-1)\gamma_A$ are nonzero integer multiples of γ_A, hence not integers. Moreover $\|x\| \in \{0, \frac12\}$ would give $2x \in \mathbb{Z}$.

(2) The inequality $A > 8\omega_3$ means $(A/\omega_3)^{1/3} > 2$, that is, $\gamma_A \in (\pi^{-1}\arctan 2, \frac12)$. Since $\tan(\pi/3) = \sqrt{3} < 2$, we have $\gamma_A > 1/3$, so $3\gamma_A \in (1, \frac32)$. Hence $\delta_{A,1} = \pi\gamma_A$ and $\delta_{A,2} = \pi(3\gamma_A - 1)$, and $\delta_{A,2} < \delta_{A,1}$ because $\gamma_A < \frac12$. As $s_{A,k}$ is nonincreasing in k, $s_{A,k} \le s_{A,2} = \delta_{A,2} < \pi\gamma_A$, and $R_A\cos s_{A,k} > R_A\cos(\pi\gamma_A) = 1$.

(3) This follows from $\sin(t-s) = 2\sin\frac{t-s}{2}\cos\frac{t-s}{2}$, $\sin t - \sin s = 2\cos\frac{t+s}{2}\sin\frac{t-s}{2}$ and $\cos\frac{t-s}{2} - \cos\frac{t+s}{2} = 2\sin\frac{s}{2}\sin\frac{t}{2}$. □

By Lemma 2.1, if γ_A is irrational and $k \geq 2$, then $F_{A,-}(s_{A,k})$ and $F_{A,+}(s_{A,k}, t_{A,k})$ are defined, and G_A is evaluated at points of $(1, \infty)$.

Definition 2.2 ([4, Definitions 7 and 8])**.** Let $A > 8\omega_3$ with γ_A irrational, and let $k \geq 2$. Condition $C(A, k)$ is

$$\max\bigl\{F_{A,-}(s_{A,k}),\, F_{A,+}(s_{A,k}, t_{A,k})\bigr\} < \delta_{A,k+1}.$$

The set $\mathbb{A}$ consists of all $A > 8\omega_3$ such that γ_A is irrational and $C(A, k)$ holds for every $k \geq 2$.

Theorem 2.3 ([4, Theorem 9])**.** *Let $C \subseteq \mathbb{R}^4$ be a symmetric convex body of revolution containing B^4 such that $|C \cap H|_3 = A$ for every hyperplane H tangent to B^4, where A does not depend on H. If $A \in \mathbb{A}$, then C is a Euclidean ball.*

2.2. **The doubled rotation number.** Put $\theta_A = 2\gamma_A$ and $\Theta = (\theta_{\min}, 1)$ with $\theta_{\min} = 2\pi^{-1} \arctan 2$. The map

$$\theta \longmapsto A(\theta) = \omega_3 \tan(\pi\theta/2)^3$$

is a diffeomorphism of Θ onto $(8\omega_3, \infty)$ with positive derivative, and its inverse is $A \mapsto \theta_A$. Since $(2k-1)\gamma_A = (k - \frac{1}{2})\theta_A$,

$$\delta_{A,k} = \pi \|(k - \tfrac{1}{2})\theta_A\| \qquad (k \geq 1). \tag{2.2}$$

We write $\Theta_{\mathrm{irr}} = \Theta \setminus \mathbb{Q}$.

Lemma 2.4. *For every $k \geq 2$, the set O_k of $\theta \in \Theta_{\mathrm{irr}}$ such that $C(A(\theta), k)$ holds is relatively open in Θ_{irr}. In particular, $\mathbb{A} = A\bigl(\bigcap_{k\geq 2} O_k\bigr)$ is a Borel set.*

Proof. The functions $\theta \mapsto \|(r - \frac{1}{2})\theta\|$ are continuous, and so are the order statistics of finitely many continuous functions. Thus $s_{A(\theta),k}$, $t_{A(\theta),k}$, $\delta_{A(\theta),k+1}$ and $R_{A(\theta)} = \sec(\pi\theta/2)$ depend continuously on $\theta \in \Theta$. On Θ_{irr}, Lemma 2.1 shows that $0 < s_{A(\theta),k} < t_{A(\theta),k} < \pi/2$ and that the arguments of $G_{A(\theta)}$ in $C(A(\theta), k)$ lie in $(1, \infty)$, where all expressions involved are continuous. Hence $\theta \mapsto \delta_{A(\theta),k+1} - \max\{F_{A(\theta),-}(s_{A(\theta),k}), F_{A(\theta),+}(s_{A(\theta),k}, t_{A(\theta),k})\}$ is continuous on Θ_{irr}, and O_k is the set where it is positive. The set Θ_{irr} is Borel and A is a homeomorphism, which gives the last assertion. □

2.3. **Continued fractions.** Let $\theta \in (0, 1)$ be irrational with continued fraction $\theta = [0; b_1, b_2, \dots]$, and let p_n/q_n be its convergents, with $p_{-1} = 1$, $q_{-1} = 0$, $p_0 = 0$, $q_0 = 1$ and

$$p_n = b_n p_{n-1} + p_{n-2}, \qquad q_n = b_n q_{n-1} + q_{n-2} \qquad (n \geq 1).$$

When the dependence on θ matters we write $b_n(\theta)$ and $q_n(\theta)$. We use the following standard facts [5]; see also [4, Propositions 10 and 11].

$$\begin{gathered} q_1 \geq q_0 = 1, \qquad q_{n+2} \geq 2q_n \quad \text{and} \quad q_n \geq 2^{\lfloor n/2 \rfloor} \qquad (n \geq 0), \\ q_{n+1} > q_n \quad \text{and} \quad q_n \leq (b_n + 1) q_{n-1} \qquad (n \geq 1); \end{gathered} \tag{2.3}$$

$$\frac{1}{q_n(b_{n+1} + 2)} < |q_n\theta - p_n| < \frac{1}{q_{n+1}} \qquad (n \geq 0); \tag{2.4}$$

$$\|m\theta\| \geq |q_n\theta - p_n| \qquad (n \geq 1,\ 1 \leq m \leq q_n). \tag{2.5}$$

For $n \geq 1$ and positive integers $c_1, \dots, c_n$, the *cylinder* $I(c_1, \dots, c_n)$ is the set of irrational $\theta \in (0, 1)$ with $b_i(\theta) = c_i$ for $1 \leq i \leq n$; we call n its rank and write $I_n(\theta)$

for the cylinder of rank n containing θ. If p_i/q_i are the convergents determined by $c_1, \dots, c_n$, the map

$$t \longmapsto \frac{p_n + p_{n-1}t}{q_n + q_{n-1}t} \tag{2.6}$$

is a bijection from the irrational points of $(0,1)$ onto $I(c_1, \dots, c_n)$; the image of $t = [0; d_1, d_2, \dots]$ has digits $c_1, \dots, c_n, d_1, d_2, \dots$, and the absolute value of the derivative of (2.6) is $(q_n + q_{n-1}t)^{-2}$. In particular, $I(c_1, \dots, c_n)$ consists of the irrational points of an interval of length $1/(q_n(q_n + q_{n-1}))$ with rational endpoints.

3. Two estimates

Lemma 3.1. *Let $A > 8\omega_3$ and $0 < s < t \le 1/4$. Then $R_A \cos s > 1$, and*

$$0 \le F_{A,-}(s) \le s^2, \qquad 0 \le F_{A,+}(s,t) \le st.$$

Proof. Write $R = R_A$ and $G = G_A$. As in the proof of Lemma 2.1, $R^2 - 1 = (A/\omega_3)^{2/3} > 4$. The function G is increasing on $[1,\infty)$ with $G(R) = 0$, and on $(1,\infty)$ its derivative $G'(x) = 1/(x\sqrt{x^2-1})$ is positive and decreasing, so G is concave. Since $s, t \le 1/4$, we have $\cos s, \cos\frac{s+t}{2} \ge 1 - \frac{1}{32} = \frac{31}{32}$, and $R^2\cos^2 s - 1 > 5 \cdot (31/32)^2 - 1 > 1$; in particular $R\cos s > 1$.

Since $R\cos s < R$, we get $F_{A,-}(s) = G(R) - G(R\cos s) \ge 0$, and by the mean value theorem

$$F_{A,-}(s) \le G'(R\cos s)\, R(1-\cos s) = \frac{1-\cos s}{\cos s\,\sqrt{R^2\cos^2 s - 1}} \le \frac{s^2/2}{31/32} < s^2.$$

Let ρ be the quotient in (2.1). Then $\rho > 1$, so $F_{A,+}(s,t) = G(R\rho) \ge 0$, and concavity gives

$$F_{A,+}(s,t) \le G'(R)\, R(\rho - 1) = \frac{\rho - 1}{\sqrt{R^2-1}} = \frac{2\sin\frac{s}{2}\sin\frac{t}{2}}{\sqrt{R^2-1}\,\cos\frac{s+t}{2}} \le \frac{st/2}{2\cdot 31/32} < st. \quad \square$$

Lemma 3.2. *Let $\theta \in (0,1)$ be irrational, and let $n \ge 1$ with $q_n \ge 2$. Then the second smallest of the numbers $\|(r - \frac{1}{2})\theta\|$, $1 \le r \le q_n$, is less than $3/q_n$. Consequently, if $A > 8\omega_3$ and $\theta = \theta_A$ is irrational, then*

$$0 < s_{A,k} < t_{A,k} < 3\pi/q_n \qquad \text{for every } k \ge q_n.$$

Proof. Write $q = q_n$, $p = p_n$, and work on the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, whose distance is $\|x - y\|$. Put $x_r = (r - \frac{1}{2})\theta$ and $y_r = rp/q - \theta/2$ for $1 \le r \le q$. By (2.4), $|\theta - p/q| < 1/(qq_{n+1})$, so

$$\|x_r - y_r\| \le r\,|\theta - p/q| < 1/q_{n+1} \qquad (1 \le r \le q).$$

Since p and q are coprime, the points $y_1, \dots, y_q$ are the points of the translated grid $q^{-1}\mathbb{Z} - \theta/2$ in $\mathbb{T}$. Every point of $\mathbb{T}$ lies within $1/(2q)$ of this grid, hence within less than $1/(2q) + 1/q_{n+1}$ of some x_r.

The points $x_1, \dots, x_q$ are pairwise distinct and different from 0 in $\mathbb{T}$, because θ is irrational. As $q \ge 2$, they divide $\mathbb{T}$ into q open arcs. Let J be the arc containing 0, and ℓ its length. The midpoint of J has distance $\ell/2$ from the nearest x_r, so $\ell < 1/q + 2/q_{n+1} < 3/q$ by (2.3). The two endpoints of J are distinct points x_r, $x_{r'}$, and each has distance less than $\ell < 3/q$ from 0. This proves the first assertion.

For the second, (2.2) gives $\delta_{A,r} = \pi\|x_r\|$. If $k \ge q$, the two smallest elements of $\{\delta_{A,1}, \dots, \delta_{A,k}\}$ are at most the two smallest elements of $\{\delta_{A,1}, \dots, \delta_{A,q}\}$, and they are distinct and positive by Lemma 2.1. $\square$

4. A tail criterion

For $j \geq 1$ put $M_j = \big((j+4)^2+2\big)^4$.

Lemma 4.1. *For every integer $j \geq 256$ we have $72M_j < 2^{\lfloor j/2 \rfloor}$.*

Proof. For $j \in \{256, 257\}$ we have $(j+4)^2 + 2 \leq 261^2 + 2 = 68123 < 2^{17}$, so $72M_j < 2^7 \cdot 2^{68} < 2^{128} = 2^{\lfloor j/2 \rfloor}$. For $j \geq 256$,

$$\frac{M_{j+2}}{M_j} = \Big(\frac{(j+6)^2+2}{(j+4)^2+2}\Big)^4 < \Big(\frac{j+6}{j+4}\Big)^8 \leq \Big(1 + \frac{1}{130}\Big)^8 < e^{8/130} < 2,$$

while $2^{\lfloor (j+2)/2 \rfloor} = 2 \cdot 2^{\lfloor j/2 \rfloor}$. Induction on j completes the proof. □

Proposition 4.2. *Let $A > 8\omega_3$ with γ_A irrational, and let b_l and q_n refer to $\theta = \theta_A$. If $b_l \leq l^2$ for every $l \geq 257$, then $C(A,k)$ holds for every $k \geq q_{256}$.*

Proof. Fix $k \geq q_{256}$. By (2.3) there is a unique $j \geq 256$ with $q_j \leq k < q_{j+1}$. By (2.3) and Lemma 4.1, $q_j \geq 2^{\lfloor j/2 \rfloor} > 72M_j \geq 72$.

Upper bound. By Lemma 3.2, $s_{A,k} < t_{A,k} < 3\pi/q_j < 1/4$. Lemma 3.1 then gives

$$\max\big\{F_{A,-}(s_{A,k}), F_{A,+}(s_{A,k}, t_{A,k})\big\} \leq t_{A,k}^2 < \frac{9\pi^2}{q_j^2}.$$

Lower bound. Put $m = 2k+1$. Then $\delta_{A,k+1} = \pi\|m\theta/2\| \geq \frac{\pi}{2}\|m\theta\|$, since $\|2y\| \leq 2\|y\|$ for every $y \in \mathbb{R}$. As $k \leq q_{j+1} - 1$, we have $m \leq 2q_{j+1} - 1$, whereas $q_{j+3} \geq q_{j+2} + q_{j+1} \geq 2q_{j+1} + q_j$. Hence $1 \leq m < q_{j+3}$, and (2.5) and (2.4) with $n = j+3$ give

$$\|m\theta\| \geq |q_{j+3}\theta - p_{j+3}| > \frac{1}{q_{j+3}(b_{j+4}+2)}.$$

By (2.3), $q_{j+3} \leq (b_{j+3}+1)(b_{j+2}+1)(b_{j+1}+1)\,q_j$. The indices $j+1, \ldots, j+4$ are at least 257, so each of the four factors $b_{j+1}+1$, $b_{j+2}+1$, $b_{j+3}+1$, $b_{j+4}+2$ is at most $(j+4)^2+2$. Therefore $q_{j+3}(b_{j+4}+2) \leq M_j q_j$ and

$$\delta_{A,k+1} > \frac{\pi}{2M_j q_j}.$$

Comparison. The inequality $9\pi^2/q_j^2 < \pi/(2M_j q_j)$ is equivalent to $q_j > 18\pi M_j$, which holds because $q_j > 72M_j$ and $18\pi < 72$. Thus $C(A,k)$ holds. □

5. Digits of quadratic growth in a cylinder

Lemma 5.1. *Let $L \geq 3$ and let I be a cylinder of rank L. Then the set*

$$E = \{\theta \in I : \ b_l(\theta) \leq l^2 \text{ for every } l > L\}$$

is a Borel set with $|E| \geq (1 - 2/L)|I|$.

Proof. Each set $\{\theta : b_l(\theta) \leq l^2\}$ is a countable union of cylinders, so E is Borel. Fix $l > L$. The cylinder I is the disjoint union of the countably many cylinders $I' \subseteq I$ of rank $l-1$. Fix such an I', let p_i/q_i be the convergents determined by its digits, and put $r = q_{l-2}/q_{l-1} \in [0,1]$. Under the bijection (2.6) with $n = l-1$, the point with parameter t satisfies $b_l = \lfloor 1/t \rfloor$, so $b_l > l^2$ holds exactly when $t < z := 1/(l^2+1)$. Hence

$$|\{\theta \in I' : b_l(\theta) > l^2\}| = \int_0^z \frac{dt}{(q_{l-1} + q_{l-2}t)^2} = \frac{z}{q_{l-1}^2(1+rz)}, \qquad |I'| = \frac{1}{q_{l-1}^2(1+r)},$$

and the ratio of these two quantities is $(1+r)z/(1+rz) \le 2z$. Summing over I' gives $|\{\theta \in I : b_l(\theta) > l^2\}| \le 2|I|/(l^2+1)$, and summing over $l > L$,

$$|I \setminus E| \le |I| \sum_{l>L} \frac{2}{l^2+1} < |I| \sum_{l>L} \frac{2}{l(l-1)} = \frac{2}{L}|I|. \qquad \square$$

6. Proof of the main results

Proof of Theorem 1.1. Let $\theta_0 = 2\gamma_0 = (4+\sqrt{2})/7$. From

$$\frac{1}{\theta_0} = 1 + \frac{2-\sqrt{2}}{2}, \qquad \frac{2}{2-\sqrt{2}} = 3 + (\sqrt{2}-1), \qquad \frac{1}{\sqrt{2}-1} = 2 + (\sqrt{2}-1)$$

we obtain $\theta_0 = [0; 1, 3, 2, 2, 2, \dots]$; in particular $b_l(\theta_0) \le l^2$ for every $l \ge 1$. By [4, Section 4.2], $A_0 = A(\theta_0)$ belongs to $\mathbb{A}$ (see also Section 7). Let W be a neighbourhood of A_0, and put $k_* = q_{256}(\theta_0)$. The parameters are chosen in the following order.

Step 1. Since $A_0 \in \mathbb{A}$, Lemma 2.4 shows that $\theta_0 \in O_k$ for each of the finitely many k with $2 \le k < k_*$, and that each such O_k is relatively open in Θ_{irr}. Hence there is an open interval V with $\theta_0 \in V \subseteq \Theta \cap A^{-1}(W)$ such that $C(A(\theta), k)$ holds for every irrational $\theta \in V$ and every $2 \le k < k_*$.

Step 2. The cylinders $I_L(\theta_0)$ are the irrational points of intervals with rational endpoints that contain the irrational number θ_0, and their lengths tend to 0. Choose $L \ge 256$ with $I_L(\theta_0) \subseteq V$.

Step 3. Let E be the set of $\theta \in I_L(\theta_0)$ with $b_l(\theta) \le l^2$ for every $l > L$. By Lemma 5.1, E is a Borel set with $|E| \ge (1-2/L)|I_L(\theta_0)| > 0$.

Step 4. Let $\theta \in E$. Then $\theta \in V \subseteq \Theta$ is irrational, so $A(\theta) > 8\omega_3$ and $\gamma_{A(\theta)} = \theta/2$ is irrational. The first $L \ge 256$ digits of θ agree with those of θ_0, so $q_{256}(\theta) = k_*$ and $b_l(\theta) = b_l(\theta_0) \le l^2$ for $257 \le l \le L$; for $l > L$ we have $b_l(\theta) \le l^2$ by the definition of E. Proposition 4.2 gives $C(A(\theta), k)$ for every $k \ge k_*$, and Step 1 gives it for $2 \le k < k_*$. Hence $A(E) \subseteq \mathbb{A} \cap W$.

Step 5. The map A is a diffeomorphism, so $A(E)$ is a Borel set, and A^{-1} is Lipschitz on compact subsets of $(8\omega_3, \infty)$, so it maps Lebesgue null sets to null sets. Since $E = A^{-1}(A(E))$ has positive measure, so does $A(E)$. $\square$

Proof of Corollary 1.2. Let $E \subseteq \mathbb{A} \cap W$ be the Borel set of positive Lebesgue measure given by Theorem 1.1. For $A \in E$ the conclusion is Theorem 2.3. $\square$

Remark 6.1. The order of the choices matters: the index 256 and the number k_* are fixed before V, and L is chosen after V. The number $k_* = q_{256}(\theta_0)$ has 98 decimal digits, but the conditions $C(A, k)$ with $2 \le k < k_*$ require no numerical work near A_0: they hold at A_0 and persist on V by continuity. The proof also shows that the relative measure of $A^{-1}(\mathbb{A})$ in the cylinder $I_L(\theta_0)$ is at least $1 - 2/L$ for all sufficiently large L.

7. The example A_0

We recall how the membership $A_0 \in \mathbb{A}$ is established in [4, Section 4.2] and describe an independent check. Let $a_0 = \pi\gamma_0$, so that $R_0 = R_{A_0} = 1/\cos a_0 \approx 2.87037$. The continued fraction of γ_0 is $[0; 2, 1, 1, 2, 2, 2, \dots]$, and the denominators of its convergents are $3, 5, 13, 31$ at the indices $2, 3, 4, 5$. Its digit of index 3 is odd and its digits of index at least 4 are even, so the denominators of index at least 2 are odd and [4, Theorem 14(a)] applies. With $j_0 = 4$, all digits of index at least 6 equal

to 2, and the monotonicity noted in [4, (21)], it reduces the conditions $C(A_0,k)$, $k\ge 7$, to the single threshold inequality

$$\frac{4\pi}{\cos(\pi/13)\sqrt{R_0^2\cos^2(\pi/31)-1}}<13. \tag{7.1}$$

It remains to verify the five initial conditions $C(A_0,k)$, $2\le k\le 6$.

Both steps reduce to signs of trigonometric expressions. Multiplying (7.1) by $\cos a_0$ and squaring positive quantities shows that (7.1) is equivalent to

$$D_{21}=169\cos^2(\pi/13)\bigl(\cos^2(\pi/31)-\cos^2 a_0\bigr)-16\pi^2\cos^2 a_0>0.$$

For the initial conditions we use the identity $G_A(x)=\arccos(1/x)-\pi\gamma_A$.

Lemma 7.1. *Let $A>8\omega_3$ with γ_A irrational, $k\ge 2$, $a=\pi\gamma_A$, $s=s_{A,k}$, $t=t_{A,k}$ and $d=\delta_{A,k+1}$.*

(1) *If $a\le d$, then $F_{A,-}(s)<d$. If $a>d$, then $F_{A,-}(s)<d$ if and only if*
$$D_-=\cos s\cos(a-d)-\cos a>0.$$
(2) *If $a+d\ge\pi/2$, then $F_{A,+}(s,t)<d$. If $a+d<\pi/2$, then $F_{A,+}(s,t)<d$ if and only if*
$$D_+=\cos a\cos\tfrac{s+t}{2}-\cos\tfrac{t-s}{2}\cos(a+d)>0.$$

Proof. (1) Since $1/(R_A\cos s)=\cos a/\cos s\in(0,1)$ by Lemma 2.1, the inequality $F_{A,-}(s)<d$ reads $\arccos(\cos a/\cos s)>a-d$. The left side is positive, which settles the case $a\le d$. If $a>d$, then $a-d\in(0,\pi/2)$, and since arccos is decreasing, the inequality is equivalent to $\cos a/\cos s<\cos(a-d)$.

(2) By (2.1), $F_{A,+}(s,t)<d$ reads $\arccos\bigl(\cos a\cos\frac{t+s}{2}/\cos\frac{t-s}{2}\bigr)<a+d$, and the argument of arccos lies in $(0,1)$, so the left side lies in $(0,\pi/2)$. This settles the case $a+d\ge\pi/2$, and otherwise the inequality is equivalent to $\cos a\cos\frac{t+s}{2}/\cos\frac{t-s}{2}>\cos(a+d)$. □

The archive `supplementary-certificate.zip` accompanying this paper contains three files: `README.md`, the program `check_constants.py`, and its output `check_constants.json`. The program runs under Python 3.12 and uses only the standard library; a successful run prints `PASS` and rewrites `check_constants.json`. All computations use intervals with rational endpoints, rounded outwards to the grid $10^{-28}\mathbb{Z}$. The program encloses $\sqrt{2}$ between rational bounds obtained from an integer square root, π by Machin's formula with alternating series remainders, and each cosine by its Taylor polynomial with the Lagrange remainder. For γ_0 it certifies the ordering of $\delta_{A_0,1},\dots,\delta_{A_0,7}$, the case distinctions of Lemma 7.1, and the signs of D_-, D_+ and D_{21}. The results for the five initial conditions are summarised in Table 1, where δ_r abbreviates $\delta_{A_0,r}$ and each number is a certified lower bound rounded down. For the threshold inequality the program certifies $D_{21}>119.186512539$. It also rechecks the integer inequalities in the proof of Lemma 4.1.

Remark 7.2. Theorem 1.1 uses the membership $A_0\in\mathbb{A}$ only through finitely many strict inequalities, namely $C(A_0,k)$ for $2\le k<k_*$; this is why a single explicit point of $\mathbb{A}$ suffices. The parity conditions of [4, Theorem 14] enter only through this point and play no role in Sections 3 to 6.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

TABLE 1. The five initial conditions $C(A_0, k)$, $2 \le k \le 6$, via Lemma 7.1.

k	$s_{A_0,k}$	$t_{A_0,k}$	test for $F_{A_0,-}$	test for $F_{A_0,+}$
2	δ_2	δ_1	$D_- > 0.120131732$	$D_+ > 0.089875327$
3	δ_3	δ_2	$D_- > 0.587760974$	$a_0 + \delta_4 > \pi/2$
4	δ_3	δ_2	$a_0 \le \delta_5$	$a_0 + \delta_5 > \pi/2$
5	δ_3	δ_2	$D_- > 0.546166802$	$a_0 + \delta_6 > \pi/2$
6	δ_3	δ_2	$D_- > 0.070272401$	$D_+ > 0.063192522$

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